Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/17/2014
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Project Title:	Epigenetic and Protein Expression Pattern Profiling of Caenorhabditis elegans Exposed to Time-Varying Gravitational Fields: a Multi-Generational Study		
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Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
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Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology (4) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	19711-5449	**Congressional District:**	1
Comments:			
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Task Description:	We hypothesize that long term temporal changes in the gravity vector will result in animal adaptation via epigenetic mechanisms that are passed onto subsequent generations. We further hypothesize that the gravity-influenced epigenetic profile can be restored to the consensus wild-type (i.e., Earth born) upon introduction to stable 1x g conditions. We propose to utilize C. elegans as a model organism to address this hypothesis by performing the following experiments. 1) Determine C. elegans behavioral responses (i.e., changes in migration dynamics, animal body posture, metabolism, etc.) due to temporal changes in the gravity vector. The gravity vector will be controlled by placing the animals inside a rotating CubeSat microscope developed by our group. The frequency, direction, and duration of rotation will be varied to elicit the most dramatic behavioral responses. Habituation time will be evaluated, and also changes in lifespan. 2) Identify gravity responsive genes by utilizing the vast collection of fluorescent strains of C. elegans available to researchers. We will monitor gene expression by fluorescence microscopy--from the embryonic stage through adulthood--during changes in the gravity vector. Genes of interest include those associated with muscle development, cell apoptosis, and neuronal development. 3) Profile the epigenetic marks on C. elegans exposed to long periods of continuous gravity vector changes that range from 1/4 of an animal's lifespan to three generations. We will examine the epigenetic consequence of restoring constant 1xg conditions over successive generations. The three Specific Aims can be conducted with the same animal strains, and thus allows the opportunity to correlate the findings across different experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding gravity's affect on biological systems is not well studied because this parameter is difficult to manipulate during experiments. The International Space Station houses the newest U.S. national laboratory, which can be utilized to understand the influence of microgravity on molecular, cellular, and animal model systems. Determining changes in epigenetic profiles during multi-generations exposed to different environments is important to know to prepare for future manned spaceflight beyond Earth. Additionally, temporary alteration of epigenomes is relevant to human health, and similar adaptation mechanisms may be displayed by almost any change in the environment or diet. In order to expedite molecular and cellular research in microgravity, we developed an innovative clinorotation microscope that enables us to observe living organisms, in real-time, during microgravity. Because of the extended clinorotation times (4 days to 3 months), we also developed a novel automated "worm chip" that permits hands-free feeding and maintenance of the animals, without interrupting clinorotation.
Task Progress:	We successfully developed an innovative clinorotation fluorescence scanning microscope. The clinorotation speed can range between 30 revolutions per minute (rpm) to over 300 rpm. We believe that this clinorotation microscope is the first of its kind because both the microscope and the specimen rotate along the same axis, which reduces image blur that would occur if only the specimen is rotated and the camera held stationary (e.g., plant clinorotation studies). The rotation speed is controlled by a motor connected to a computer, and rotation speed is measured by optical encoders. Preliminary tests show that the instrument can operate reliably over four days (rotation speed is maintained with no apparent overheating of the motor). Currently, we are collecting data with metal nanoparticles and micro-spheres to calibrate the instrument for simulated microgravity. The worm community has grown and maintained C. elegans on E. coli seeded NGM (agar) plates since the 1970s. However, this method of culturing requires continued monitoring and periodic hand transfer of worms from one plate to another. It is often difficult to avoid bacterial and yeast contamination during genomic, biochemical, and microscopic studies. To reduce the risks of contamination, we devised a fully automated system to culture C. elegans in a chemically defined, axenic medium (CeMM), similar to the liquid media used to maintain worms previously flown on the International Space Station. We housed C. elegans in devices constructed out of gas-permeable polydimethylsiloxane (PDMS). These devices allow for liquid CeMM and animals to be introduced into a sealed 1 mL volume chamber (16mm diameter × 5mm thickness), while permitting oxygen to diffuse in and out of the chambers and inhibiting the formation of air bubbles. Bubble-free liquid media is required for successful clinorotation, to avoid any shear forces caused by moving air bubbles. A programmable peristaltic pump connected to tubing at two ends of the chamber enables precise control of transfer of media and worms through the device. Currently, we exchange a total of 200 µl of liquid per day. In approximately three weeks, around 100,000 animals will propagate in a single device, allowing for enough DNA to perform epigenetic screenings (ChIP-Seq). We are also performing experiments to maintain a steady-state number of healthy worms in each chip, by periodically flushing 10-20% of the worms to avoid overcrowding. We characterized phenotypic variations of animals grown in liquid cultures in our worm chips and glass vials, and compared these variations to animals grown on conventional agar plates. We found no discernible changes in growth between the glass vials and worm chips. We observed that the animals grown in liquid displayed an initial adaptation to the environment, taking ~10-12 days to develop into adult as opposed to 4-5 days on agar. Animals of the following generations took 6-8 days to develop into adults. The liquid animals were also slightly thinner and longer compared to those grown on agar. The most significant difference we observed was a dramatic decrease in fecundity. On agar, worms produced 250-300 progeny, compared to 15-20 progeny in liquid media. The major difference between the worms in liquid and agar environments is the presence of adhesion forces between the worms and the agar. The phenotypic differences between worms grown on agar and liquid cultures have been observed and published by many groups. However, the cellular mechanism(s) responsible for a significantly low broad size has not been published. To examine the cause of this reduction in broad size, we dissected out the germlines of early adult wild-type hermaphrodites grown in liquid and agar plates, and stained the gonads with DAPI (a fluorescent dye that binds to DNA) to examine nuclear morphology. We found that animals grown on agar display normal germ cell progressions from distal to proximal regions of the germline, whereas the gonads of worms in liquid were overall reduced in size and the cellular morphologies were severely deformed. Markedly, the mitotic zones were reduced to ~7 to 10 cell-diameters and the overall width of the gonads was reduced at least two-fold. Cells in meiotic development (pachytene region) display spatial disorientation of germ cells, leading to fewer oocytes and few sperms in the proximal region. We have also recently acquired animal strains that contain GFP-Histone2B fusions to repeat experiments with an intrinsic germline fluorescent marker in live animals, and use these strains in clinorotation studies. We speculate that the changes in the germlines between the liquid and agar cultures have not been previously observed because we are utilizing two fairly exotic experimental techniques: 1) C. elegans are not typically maintained for several generations in liquid media, and 2) dissecting the C. elegans germline for fluorescence imaging is technically challenging and not many labs are trained in this procedure. We are very confident about the germline morphologies, and have repeated the experiments a number of times, including liquid CeMM controls in glass vials to overrule any adverse

	chemical effects from the PDMS. An exciting milestone in this project will be epigenetic profiling of C. elegans in liquid media and agar. The epigenetic profiles will reveal the liquid adaptation mechanisms at the genomic level, which we will correlate with germline morphology (by identifying germline-associated epigenetic silencing/expression). The epigenetic profile of the liquid cultures will provide the baseline for clinorotation studies. Chromatin Immuno-Precipitation followed by Next Generation DNA sequencing (ChIP-Seq) is currently underway. We have “ChIP-ed” five target histone methylation/acetylation marks, and RNA polymerase as a control. ChIP sequencing libraries have recently been made, and we anticipate the first epigenetic results by mid-August. Having optimized all the necessary experimental procedures for epigenetics, the rate limiting step is now the cultivation of worms, i.e., harvesting enough DNA for sequencing. We are currently working on two publications: one describing automatic culturing of C. elegans, and the other describing a novel clinorotation microscope with data from bead standards (comparing theory with experiments). Drs. Sabanayagam, Doh, and Jayasinghe (ASTID postdoctoral researcher) have submitted two abstracts to the 2014 American Society for Gravitational and Space Research meeting, based on the above publications in preparation. Also, we are pressing hard to obtain the epigenetic data and begin preparing a manuscript on germline aberrations observed in liquid cultures, with a target submission date of December 2014.
Bibliography Type:	Description: (Last Updated: 08/17/2023)
Significant Media Coverage	Biophysical Society. "Dr. Sabanayagam's NASA research highlighted in Biophysical Society's 2014 Summer Research Program in Biophysics brochure." Biophysical Society 2014 Summer Research Program in Biophysics brochure, May 2014., May-2014